

Web Development

>> HTML: Forms

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HTML <form> Elements

Tag	Description
<form>	Defines an HTML form for user input
<input>	Defines an input control
<textarea>	Defines a multiline input control (text area)
<label>	Defines a label for an <input> element
<fieldset>	Groups related elements in a form
<legend>	Defines a caption for a <fieldset> element
<select>	Defines a drop-down list
<optgroup>	Defines a group of related options in a drop-down list
<option>	Defines an option in a drop-down list
<button>	Defines a clickable button
<datalist>	Specifies a list of pre-defined options for input controls
<output>	Defines the result of a calculation

The <label> Element



- Defines a label for several form elements.
- The for attribute of the <label> tag should be equal to the id attribute of the <input> element to **bind them together**.
- Example:

```
<label for="fname">First name:</label>  
<input type="text" id="fname" name="fname">
```

The <fieldset> Element

- Used to group related elements in a form.
- Draws a box around the related elements.
- <legend> tag is used to define a caption for the <fieldset> element.

```
<form action="/action_page.php">
  <fieldset>
    <legend>Personal information:</legend>
    <label for="fname">First name:</label>
    <input type="text" id="fname" name="fname"><br><br>
    <label for="lname">Last name:</label>
    <input type="text" id="lname" name="lname"><br><br>
    <label for="email">Email:</label>
    <input type="email" id="email" name="email"><br><br>
    <label for="birthday">Birthday:</label>
    <input type="date" id="birthday" name="birthday"><br><br>
    <input type="submit" value="Submit">
  </fieldset>
</form>
```

The fieldset element

Personal information:

First name:

Last name:

Email:

Birthday: 

The <select> Element

- Defines a drop-down list <select>
- <option> elements defines an option that can be selected.

```
<!DOCTYPE html>
<html>
<body>

<form action="/action_page.php">
  <label for="cars">Choose a car:</label>
  <select id="cars" name="cars">
    <option value="volvo">Volvo</option>
    <option value="saab">Saab</option>
    <option value="fiat">Fiat</option>
    <option value="audi">Audi</option>
  </select>
  <input type="submit">
</form>

</body>
</html>
```

<option value="fiat
" selected>Saab
</option>

Choose a car:

Volvo
Saab
Fiat
Audi

The <datalist> Element

- The **<datalist>** element presents a list of suggested values to the associated field and users are free to *select one of those suggested values* or *type in their own value*.

`<input list="">`
`<datalist`
`id="">`

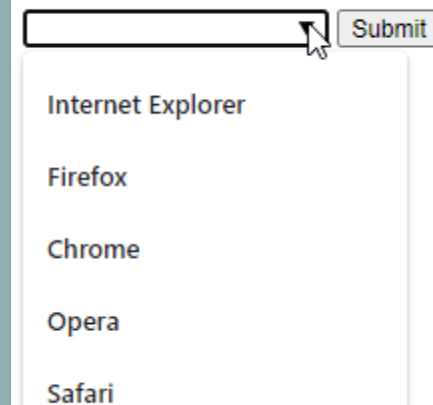
```
<!DOCTYPE html>
<html>
<body>

<h2>The datalist Element</h2>

<form action="/action_page.php">
  <input list="browsers" name="browser">
  <datalist id="browsers">
    <option value="Internet Explorer">
    <option value="Firefox">
    <option value="Chrome">
    <option value="Opera">
    <option value="Safari">
  </datalist>
  <input type="submit">
</form>

</body>
</html>
```

The datalist Element



Internet Explorer

Firefox

Chrome

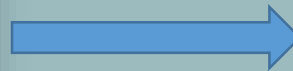
Opera

Safari

The <optgroup> Element

- Used to **group** related options in the **dropdown list**

```
<form action="/action_page.php">
  <label for="cars">Choose a car:</label>
  <select name="cars" id="cars">
    <optgroup label="Swedish Cars">
      <option value="volvo">Volvo</option>
      <option value="saab">Saab</option>
    </optgroup>
    <optgroup label="German Cars">
      <option value="mercedes">Mercedes</option>
      <option value="audi">Audi</option>
    </optgroup>
  </select>
  <br><br>
  <input type="submit" value="Submit">
</form>
```



The optgroup element

The optgroup tag is used to group related options in a drop-down list:

Choose a car:

- Swedish Cars
 - Volvo
 - Saab
- German Cars
 - Mercedes
 - Audi



The `<input>` Element

Input Types



<code><input type="text"></code>	<code><input type="image"></code>
<code><input type="checkbox"></code>	<code><input type="month"></code>
<code><input type="button"></code>	<code><input type="number"></code>
<code><input type="date"></code>	<code><input type="password"></code>
<code><input type="datetime-local"></code>	<code><input type="radio"></code>
<code><input type="email"></code>	<code><input type="search"></code>
<code><input type="file"></code>	<code><input type="submit"></code>
<code><input type="hidden"></code>	<code><input type="url"></code>

Radio

- Radio buttons let a user select **ONLY ONE** of a limited number of choices.

```
<input type="radio" id="red" name="color" value="red" />  
<label for="red">Red</label><br />  
<input type="radio" id="green" name="color" value="green" />  
<label for="green">Green</label><br />  
<input type="radio" id="blue" name="color" value="blue" />  
<label for="blue">Blue</label>
```



☐ Red
☒ Green
☐ Blue

Multiple radio button groups in one form

- Using `<fieldset>` element, link **name** attribute of each related **radio** input to the **id** of the **fieldset** it belongs to.

```
<form>
  <fieldset id="group1">
    <input type="radio" name="group1">value1</input>
    <input type="radio" name="group1">value2</input>
  </fieldset>

  <fieldset id="group2">
    <input type="radio" name="group2">value1</input>
    <input type="radio" name="group2">value2</input>
    <input type="radio" name="group2">value3</input>
  </fieldset>
</form>
```

Checkbox

- Checkboxes let a user select **ZERO** or **MORE** options of a limited number of choices.

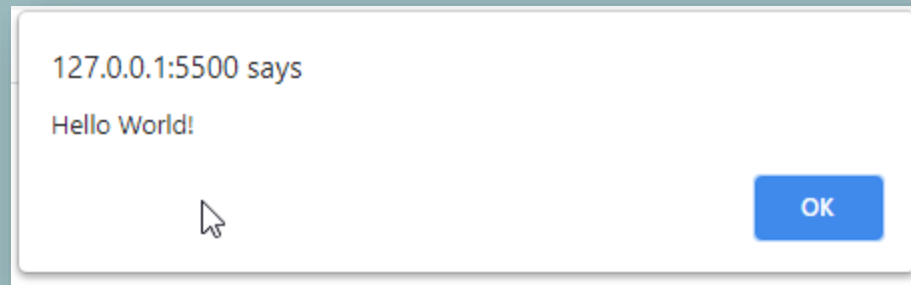
```
<input type="checkbox" id="course1" name="course1" value="Genomes" />  
<label for="course1"> Genome Analysis</label><br />  
<input type="checkbox" id="course2" name="course2" value="Network" />  
<label for="course2"> Networks</label><br />  
<input type="checkbox" id="course3" name="course3" value="Statistics" />  
<label for="course3"> Biostatistics</label>
```



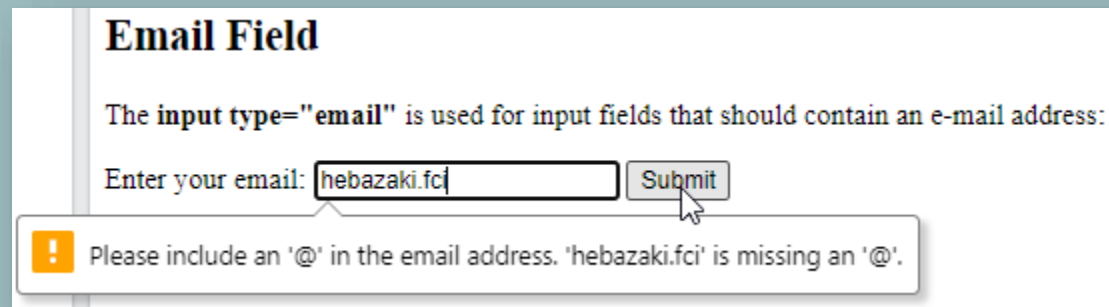
☒ Genome Analysis
☐ Networks
☒ Biostatistics

Button And Email

- `<input type="button" onclick="alert('Hello World!')" value="Alert!">`



- The e-mail address can be automatically validated when submitted.
- `<input type="email" id="email" name="email">`



<Input> Restrictions

Attribute	Description
checked	Specifies that an input field should be pre-selected when the page loads (for type=" checkbox " or type=" radio ")
disabled	Specifies that an input field should be disabled
max	Specifies the maximum value for an input field
maxlength	Specifies the maximum number of character for an input field
min	Specifies the minimum value for an input field
pattern	Specifies a regular expression to check the input value against
readonly	Specifies that an input field is read only (cannot be changed)
required	Specifies that an input field is required (must be filled out)
size	Specifies the width (in characters) of an input field (text, search, tel, url, email, and password)
value	Specifies the default or an initial value for an input field

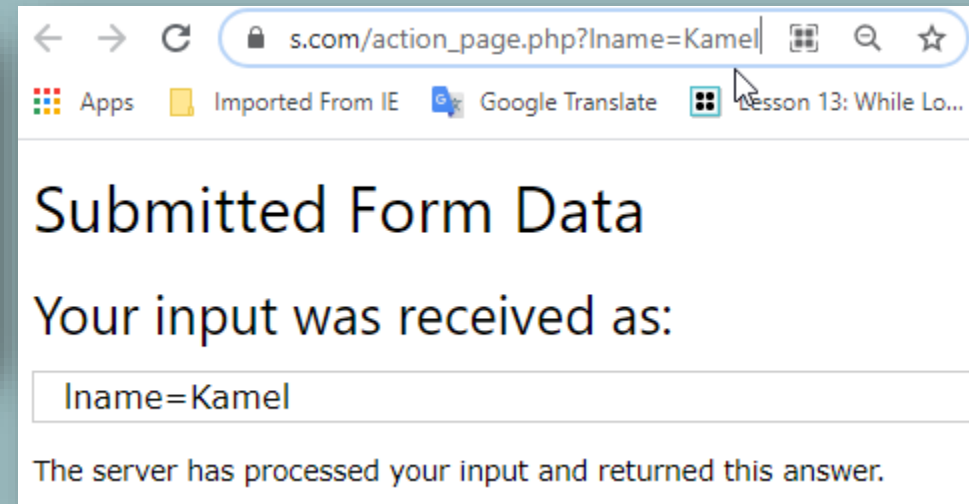
Examples

- `<input type="number" id="quantity" name="quantity" min="0" max="100" value="30">`
- `<input type="range" id="vol" name="vol" min="0" max="50">`
- `<input type="text" id="country_code" name="country_code" pattern="[A-Za-z]{3}" title="Three letter country code">`
- `<input type="tel" id="phone" name="phone" placeholder="123-45-678" pattern="[0-9]{3}-[0-9]{2}-[0-9]{3}">`
- `<input type="text" id="username" name="username" required>`

Name attribute for <input>

- Each **input** field must have a **name** attribute to be submitted.
- If the **name** attribute is omitted, the **value** of the **input** field will not be sent at all.

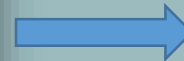
```
<form action="/action_page.php" target="_blank" method="get">
  <label for="fname">First name:</label><br>
  <input type="text" id="fname" value="Ahmed"><br>
  <label for="lname">Last name:</label><br>
  <input type="text" id="lname" name="lname" value="Kamel"><br>
<br>
  <input type="submit" value="Submit">
</form>
```



list attribute for <input>

- The **list** attribute of the **<input>** element, must refer to the **id** attribute of the **<datalist>** element.

```
<p style="color: ■blue">  
  This is list of hobbies please, select one of them:  
</p>  
  
<input list="hobbies" />  
<datalist id="hobbies">  
  <option value="Reading"></option>  
  <option value="Swimming"></option>  
  <option value="Football"></option>  
  <option value="Music"></option>  
  <option value="Social media"></option>  
</datalist>
```



This is list of hobbies please, select one of them:

Reading
Swimming
Football
Music
Social media

Thank you